A Supplementary material

A.1 Coding process for the *akim*s dataset

After extracting each line of akims' job histories, I code a summary of the job, the type/'rank' (a more generalized summary), the organization, and the place of the job (in the 1997–2018 oblast' boundaries). About two-thirds of the first three variables can be coded by string matching because they use regular phrases to refer to jobs which always have the same employer. For example, common phrases coded in this way include *ministr* (minister, central government) or *polnomochnyy posol* (ambassador, central government). Some are only coded for type—for example, entries for *rukovoditel' apparata akima* refer to the head of an akim's apparatus (equivalent to the president's chief of staff), but whether that is at the oblast' or rayon level needs manual coding. Finally, I manually code all job locations. Examples of this coding process are in Table A.1, which includes ten randomly sampled examples from the data.

In some contexts—such as China—using cadres' job titles to identify rank is relatively simple. In Kazakhstan, the cadre system is less well defined. Context is as important as job title. Reappointment from a relatively poor region to lead a ministry might clearly be a promotion where the same move from a rich region would not be. These data can only provide a rough measure of rank. I code recorded positions into a number of categories. For instance, a 'minister' is the political head of a ministry, such as the minister of finance.[A1] A 'department head' is a ministry's senior civil servant, and a 'department committee head' the person who leads a ministerial subdepartment, agency, or committee. In the ministry of finance, one such body is the Agency for Customs Control. A 'department regional branch head' has a specific territorial responsibility (usually defined by oblast') but works under the aegis of a central department, rather than being responsible to an akim. The Ministry of Finance, for example, maintains oblast' Committees for Financial Control to audit regional spending. Similar divisions exist in governor's offices and their departments. Other individuals, for example, are coded as 'political advisor' if they work directly for a senior political figure (in most cases, the president or prime minister).

I code some entries—especially for the location variable—as 'unknown' because the entry is unclear. Often, this is for early, Soviet-era work where an individual is recorded as a 'worker' in an unspecified *sovkhoz* (state farm) or similar. A potentially bigger problem is entries, concentrated in the 1990s, involving work as a director for a *TOO*

[A1] I code 'Prime Minister' and 'Vice Prime Minister' as Minister and Vice Minister respectively.

Table A.1: Examples of biography entry coding

Original	Translation	Type	Organisation	Place
Председатель Ленинского райисполкома города Алма-Аты (1986-1989)	Chair of the Leninskiy rayon executive committee of Alma-Ata city	District government leadership (Soviet)	District government (Soviet)	Almaty city
Советник-посланник Посольства Республики Казахстан в Исламской Республике Иран (2007-2008)	Advisor-envoy to the Ambassador of the Republic of Kazakhstan to Iran	Consul	Central government	Rest of World
Советник председателя совета директоров - Генеральный представитель АО «Народный банк Казахстана» (11.2013-02.2014)	Advisor to the Chair of the Board — General Representative of AO "People's Bank of Kazakhstan"	Private firm leadership	Private business	Almaty city
Начальник Каражанбасского управления по повышению нефтеотдачи пластов и капитальному ремонту скважин и начальником нефтегазодобывающего управления «Каражанбастермнефть» (1982-1987)	Head of the Karazhanbasskiy Department for Enhanced Oil Recovery and Well Overhaul and Head of "Karazhanbastermneft" Department for Oil and Gas Production	Soviet management	Soviet agriculture, mining, and enterprise	Mangistau
Заместитель руководителя Администрации Президента Республики Казахстан (04.1998-08.1998)	Vice Head of the Administration of the President of the Republic of Kazakhstan	Department vice head	Central government	Astana
Директор Уральского филиала ОАО «Темирбанк» (1998-2003)	Director of the Ural'sk brank of OAO "Temirbank"	Worker	Private business	ZKO
Преподаватель Актюбинского государственного педагогического института, директор Центра подготовки олимпийского резерва Актюбинской области (2004-2007)	Chair of the Aktobe State Pedagoloichical Institute, Director of the Training Centre for the Olympic Reserve of Aktobe Oblast'	Research/University administration	Academia	Aktobe
Заместитель Премьер-Министра Республики Казахстан (25.02.2019-20.08.2019)	Vice Prime-Minister of the Republic of Kazakhstan	Vice minister	Central government	Astana
Исполняющий обязанности директора, директор департамента финансового регулирования Министерства транспорта и коммуникаций Республики Казахстан (2002-2003)	Acting director, director of the Department of Financial Regulation of the Ministry of Transport and Communication of the Republic of Kazakhstan	Department committee head	Central government	Astana
Электромонтер станции Кандагач (1984)	Electrician, Kandahach station	Worker	District government (Soviet)	Aktobe

(equivalent to a western LLP), or similar business, that has long since stopped operating and has no internet presence. As well as this usually meaning place of work cannot be determined, the difficulty here is that businesses like these can be private concerns, run by local government, or part of national SOEs. I record these as 'unknown' rather than making any assumptions about individuals' employers. Some coding involves a degree of conjecture. The most important example is for ministerial and other government positions. Location for these is generally not explicitly given, but it is reasonable to assume that ministers work in the capital. Inaccuracy here may be introduced as a result of the 1997 switch of Kazakhstan's capital from Almaty city to Astana. I code these positions as being in Astana from the end of 1997, though some ministries may not have completed the move until after this date. Nevertheless, this issue only affects a small proportion of entries. Moreover, these are only used in descriptive analyses where, for this period, both Almaty and Astana can be thought of as 'the center'.

In Table A.2, I summarize this coding. By far the most common type of job recorded in biographies is work for central or regional government. This includes work as an akim (or regional administration head, as the position was known in the early 1990s).[A2] It also includes a variety of senior-level positions in government structures, as minister/vice minister, vice akim, or the bureaucratic head of a department or departmental committee. Given this, it is unsurprising that the most common places of work recorded in the biographies are Astana and Almaty, respectively the country's capital and biggest city (and pre-1997 capital). Other regions are far less frequently recorded in the data. For example, more biography entries record work abroad, in an embassy or at an international organization, than in eight regions. This reflects the importance of work in the center, as well as the concentration of people—and political opportunities—in certain parts of Kazakhstan.

A.1.1 Promotion and demotion coding

Finally, in line with the discussion in the data section, I code almost all positions as a 'promotion' or 'demotion' as follows:

> Promotion (66 cases). A plausibly more connected/central state role than the governor's current regional leadership position. For example, governor of a somewhat bigger region, more minor ministerial roles (e.g., minister for tourism), non-head leadership in a significant national SOE, or to

[A2] Note that the count of jobs as an akim is higher than the number of akim tenures included in my core analyses because this career path data includes the period prior to 1997 (i.e., an akim may have started their tenure in my data in 2000, but have previously held an akim post in 1995, which is included in the 'pre-tenure' career data but not as a tenure).

Table A.2: Summary of the career path data

Job type	Count	Organisation	Count	Place	Count
Oblast akim/Head of Regional Administration	166	Central government	570	Astana	538
SOE leadership	117	Regional government	312	Almaty city	218
Vice minister	111	Soviet agriculture, mining, and enterprise	129	Unknown	126
Minister	97	SOE	119	Aktobe	68
Worker	90	District government	103	ZKO	65
Oblast vice akim/Deputy Head of Regional Administration	85	Unknown	76	YuKO/ Shymkent/ Turkestan	62
Department committee head	84	District government (Soviet)	54	Karaganda	60
Rayon/GOZ akim/Head of District Administration	75	Academia	45	Pavlodar	60
Unknown	74	Private business	43	Kostanay	51
Soviet functionary	68	Regional government (Soviet)	39	Akmola	47
Engineer/geologist	48	Komsomol	32	Almaty	46
Department head	45	Central government (Soviet)	22	Atyrau	43
Regional government leadership	43	Representative body	19	SKO	42
Political advisor	40	Quango	16	VKO	41
Private firm leadership	38	Ruling party	16	Mangistau	39
Ambassador	37	Small business and agriculture	15	Kyzylorda	35
Department functionary	36	Local government (Soviet)	10	Zhambyl	34
District government leadership (Soviet)	34	International organisation	6	Russian Federation	18
Department regional branch head	32	Local government	6	FSU (exluding Russia)	17
Department vice head	30	Retired	5	Rest of World	17
Other	295	Other	8	Other	18
Total	1645	Total	1645	Total	1645

Note: Data are individual level—i.e., each biography is only counted once, not once per tenure. Soviet and non-Soviet government work is separated—e.g., Obkom first secretary is recorded as a separate type of work to akim/regional administration head and catagorised under 'Regional government (Soviet)'. Includes full reported career information for 113 individuals.

lead a national SOE, diplomatic postings to large countries and regional neighbors. Includes 'clear promotions'.

Clear promotion (39 cases). A clearly more important state position. For example, important minister, governor of a region with 150% or more GRP/the capital, ambassador to Russia.

Demotion (14 cases). A somewhat worse state role than the governor's current regional leadership position. E.g., slightly smaller region, advisory role in the presidential administration, leadership role in a minor SOE. Also includes retirement or leaving the civil service to take a senior private sector role. Includes 'clear demotions'.

Clear demotion (7 cases). A clearly worse state role, prosecution, or departure from the civil service without taking a private sector role. For example, vice governor of a smaller region, regional head of a minor national structure, minor international role (e.g., 'head of the Kazakh-Belarusian cultural exchange agency').

So, 26 tenures end with a role that I code as neither a demotion nor a promotion.

A.2 Data

A.2.1 Turnover

Figure A.1 presents more detail on the dependent variable—the timing of oblast' akims' dismissals—showing the month they left office for the full length of my data. The left panel is a set of timelines by oblast', with dots indicating dismissal. The right panel is a histogram of the count of dismissals by month (i.e., how many akims were dismissed in month x).

A.2.2 Nightlights data

The main nightlights data are taken from Chen et al. (2026), who provide an annual, pre-background zeroed, harmonized series of VIIRS-like radiance from 1992–2025 (here, I use data from 2002–2021 to match the other economic statistics). Until 2013, these data are model predictions from older, lower-resolution sensors (DMSP-OLS) calibrated to match the features of newer VIIRS data. The data are taken directly from the paper's dataverse (`doi:10.7910/DVN/YGIVCD`). I do not mask the data to account

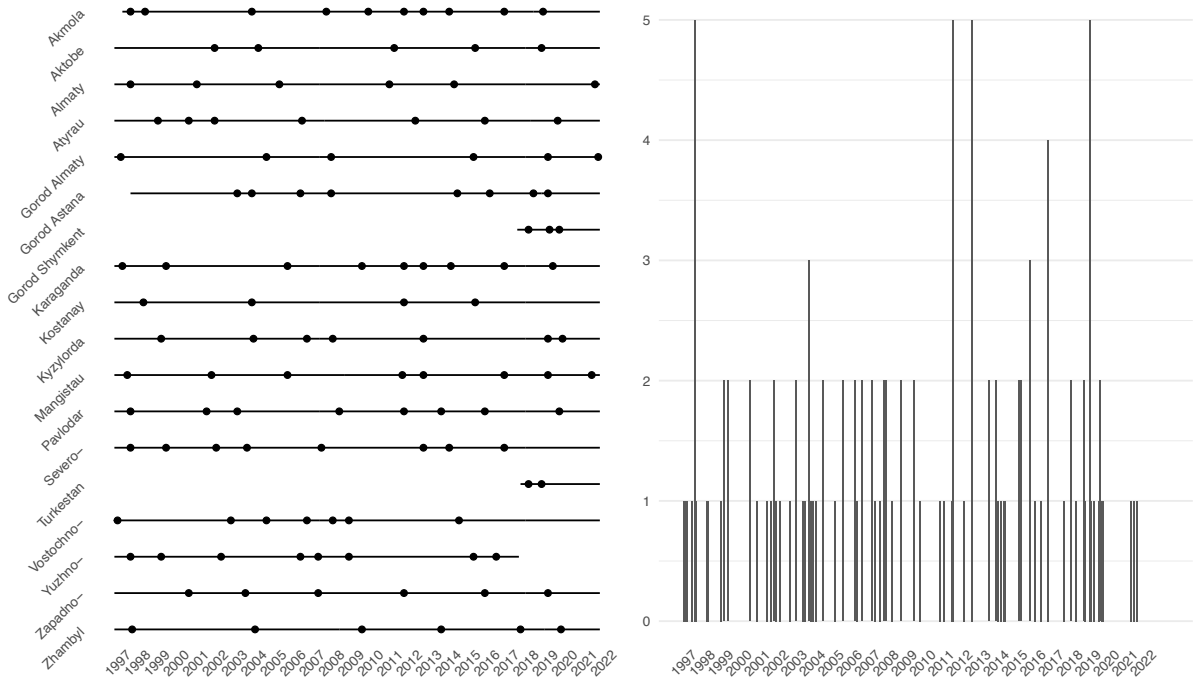


Figure A.1: Departures by oblast' and monthly departure count

for flaring (flaring site data are only available for 2012–2019); I note that accounting for flaring results in an essentially unchanged growth measure anyway (correlation $r > 0.99$). Finally, I drop the 2013 observations to exclude the switch between two series in Chen et al. (2026) which results in incomparable radiance measures.

I match the series with Kazakhstan's 2018 region boundaries from Rodionov (2021). Note that (a) Shymkent is included in the South Kazakhstan measure even after it leaves the oblast' in 2018, (b) Almaty city received over 200 km^2 from Almaty oblast' in 2014, which us therefore included as in the city for the full dataset, and (c) Baikonur (leased by the Russian Federation) is included in Kyzylorda oblast' throughout. I calculate flux as radiance integrated over the region (using `exactextractr`; where cells overlap at regional borders, they are apportioned by coverage fraction). The final growth measure is then $100 \times \Delta \log(\text{flux})$. In the tests, this growth variable is then demeaned within years and lagged by one year.

I provide two robustness checks to assuage concerns about the limitations of the harmonized data. Beyond dropping the 2013 observation, the Chen et al. (2026) series is annual so I cannot drop observations contaminated by snow (which overstates lighting and is structurally biased toward the north of the country). As a robustness check, I therefore also employ a monthly NOAA-supplied cloud-free VIIRS product, which is only available from 2012 (taken from the World Bank's `globalnightlight` bucket). I

create an annual composite of the median of April, August, September, and October. Here, I use flare masking of Earth Observation Group (EOG) designated oil and gas sites (as from 2012–2019). There is no pre-processing (zeroing) of noise, so I calculate the noise floor from radiance over cells with no population within roughly 5 km. While growth calculated from these better data are only correlated at 0.468, the results using it are very similar.

Second, I additionally test both series' mean radiance weighted by population. This accounts for different population patterns across the country as well as some regions appearing well 'lit' due to oil and gas flaring over empty desert. I weight using WorldPop's 2020 constrained 100m grid (summed to the light grid).

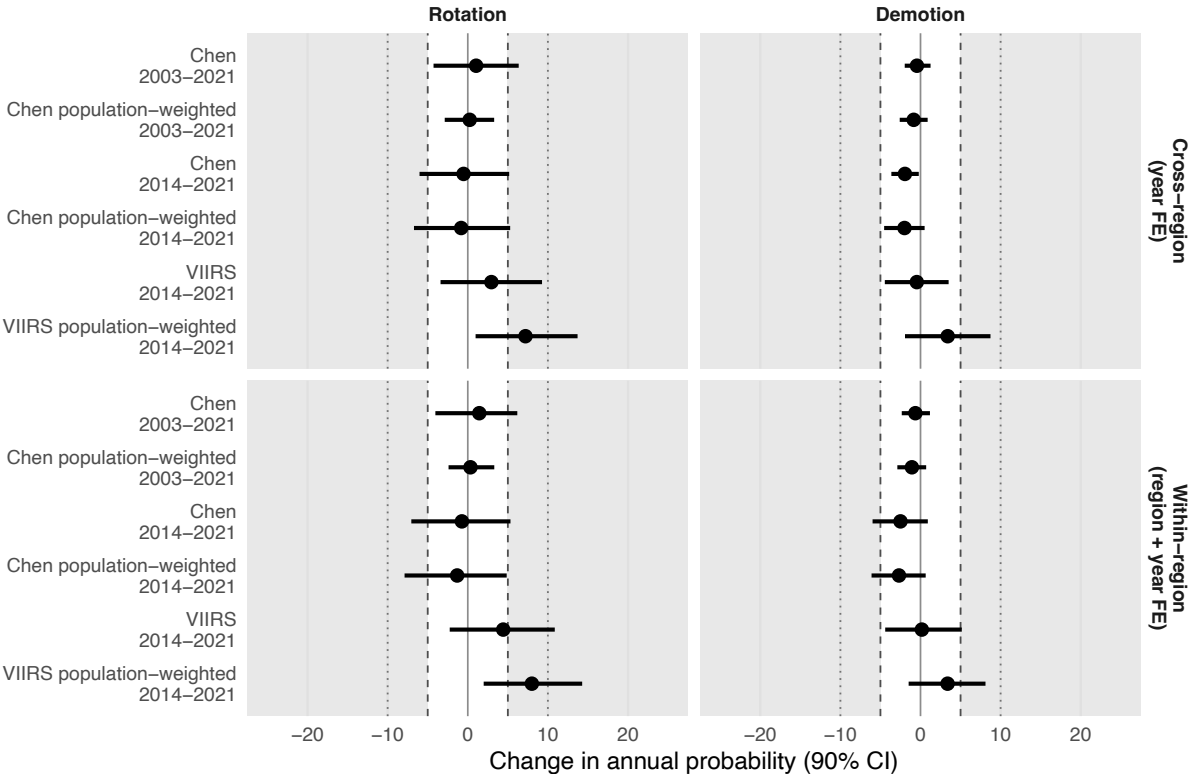


Figure A.2: Equivalence tests of performance indicators on governors' probability of rotation and dismissal with various nightlight measures

A.3 Robustness and sensitivity checks

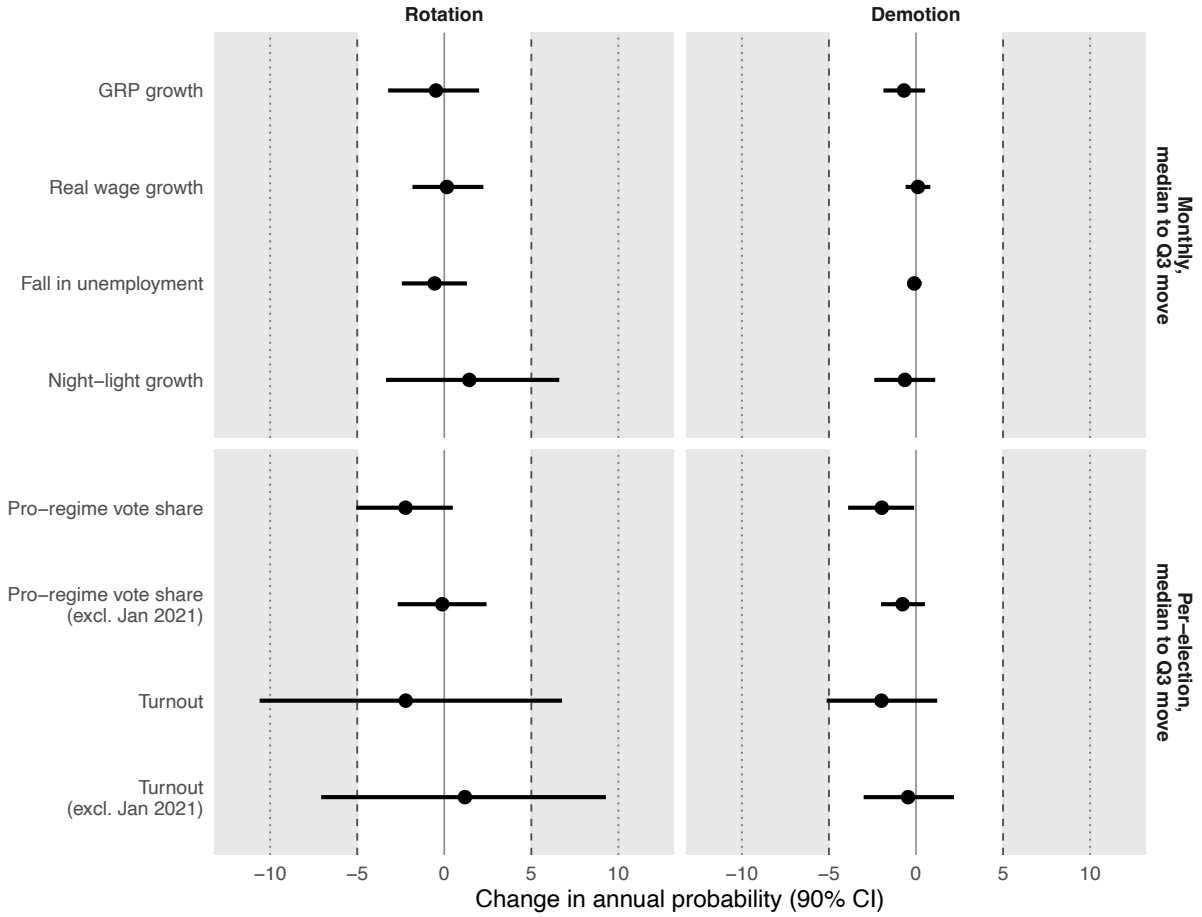
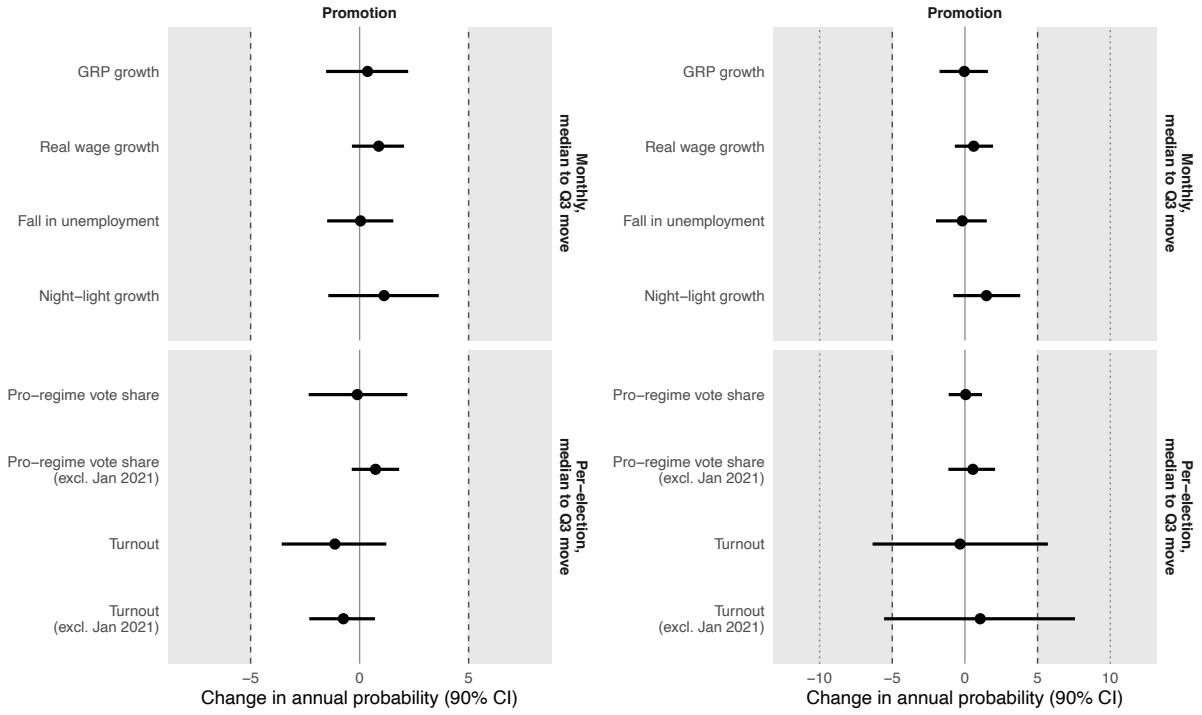


Figure A.3: Equivalence tests of performance indicators on governors' probability of rotation and dismissal, including regional fixed effects



(a) Cross-region comparison

(b) With regional fixed effects

Figure A.4: Equivalence tests of performance indicators on governors' probability of promotion